6G USING MEAN BOND ENTHALPIES

SPECIFICATION REFERENCE: 6.8 6.9 6.10 6.11

LEARNING OBJECTIVES

- Be able to use bond enthalpies to calculate enthalpy changes, understanding the limitations of this method.
- Be able to calculate mean bond enthalpies from enthalpy changes of reaction.
- Understand that bond enthalpy data gives some indication about which bond will break first in a reaction, how easy or difficult it is, and therefore how rapidly a reaction will take place.
- Be able to define standard enthalpy change of atomisation.

CALCULATING AN ENTHALPY CHANGE OF REACTION USING MEAN BOND ENTHALPIES

Here is the process for calculating an enthalpy change of reaction using mean bond enthalpies.

- Step 1: Calculate the sum of the mean bond enthalpies of the bonds broken, Σ(bonds broken).
- Step 2: Calculate the sum of the mean bond enthalpies of the bonds made, Σ(bonds made).
- Step 3: Calculate the enthalpy change of reaction using the equation:

$$\Delta_r H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds made})$$

WORKED EXAMPLE 1

Calculate the enthalpy change of reaction for:

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

given the following data:

E(H–H) = 436 kJ mol^{-1}

E(Cl–Cl) = 244 kJ mol^{-1}

E(H–Cl) = 432 kJ mol^{-1}

Answer

Σ(bonds broken) = (436 + 244) = 680 kJ mol^{-1}

Σ(bonds made) = (432 × 2) = 864 kJ mol^{-1}

$\Delta_r H$ = (680 – 864) = –184 kJ mol^{-1}

WORKED EXAMPLE 2

Calculate the enthalpy change of reaction for:

$$H_2O_2(l) \rightarrow H_2O(l) + \tfrac{1}{2}O_2(g)$$

given the following data:

E(O–H) = 463 kJ mol^{-1}

E(O–O) = 146 kJ mol^{-1}

E(O=O) = 496 kJ mol^{-1}

If you look at the displayed formula for each substance you will see that the two O–H bonds in hydrogen peroxide, H_2O_2, are also present in water:

H–O–O–H and H–O–H

So we do not have to include these in our calculations.

Answer

Σ(bonds broken) = 146 kJ mol^{-1}

Σ(bonds made) = $\frac{1}{2}$ × 496 kJ mol^{-1} = 248 kJ mol^{-1}

$\Delta_r H$ = (146 – 248) = –102 kJ mol^{-1}

LEARNING TIP

If you are unsure which bonds break in the reaction, calculate the sum of the bond enthalpies for all of the bonds in both the reactants and products, as in worked example 1.

LIMITATIONS OF THIS METHOD OF CALCULATION

The measured value for the enthalpy change of this reaction is $-98\,kJ\,mol^{-1}$. The reason for the difference is that bond enthalpies are measured in the gaseous state, and both hydrogen peroxide and water are liquids in the reaction. Also, mean bond enthalpies have been used and these may not correspond to the bond enthalpies in the molecules themselves.

CALCULATING MEAN BOND ENTHALPIES FROM ENTHALPY CHANGES OF REACTION

For this type of calculation, you will be supplied with a value for the enthalpy change of a reaction, together with all the relevant mean bond enthalpies except one: the one you are asked to calculate.

To solve the problem, simply substitute the known mean bond enthalpies and the unknown bond enthalpy into the expression:

$$\Delta_r H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds made})$$

Rearrange the expression to make the unknown bond enthalpy the subject, and solve the problem.

WORKED EXAMPLE 3

Calculate the bond enthalpy of the C=C bond in ethene, given the following data.

$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$ $\quad \Delta_r H = -147\,kJ\,mol^{-1}$

$E(H{-}H) = 436\,kJ\,mol^{-1}$

$E(C{-}H) = 413\,kJ\,mol^{-1}$

$E(C{-}C) = 347\,kJ\,mol^{-1}$

$\Sigma(\text{bonds broken}) = E(C{=}C) + E(H{-}H) = E(C{=}C) + 436\,kJ\,mol^{-1}$

$\Sigma(\text{bonds made}) = 2E(C{-}H) + E(C{-}C) = (2 \times 413) + 347 = 1173\,kJ\,mol^{-1}$

$\Delta_r H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds made})$

$-147 = E(C{=}C) + 436 - 1173$

$E(C{=}C) = 1173 - 147 - 436$

$E(C{=}C) = 590\,kJ\,mol^{-1}$

WORKED EXAMPLE 4

In the following example, carbon is in the solid state in the reaction, so you need to think about the enthalpy change for the conversion of the solid into the gas.

Calculate the mean bond enthalpy of the C–H bond in methane using the following data.

$\Delta_f H^{\ominus}[CH_4(g)] = -75\,kJ\,mol^{-1}$

$C(s, \text{graphite}) \rightarrow C(g)$

$\Delta H^{\ominus} = +715\,kJ\,mol^{-1}$

$E(H{-}H) = 436\,kJ\,mol^{-1}$

The equation that represents the standard enthalpy change of formation of methane is:

$C(s, \text{graphite}) + 2H_2(g) \rightarrow CH_4(g)$ $\quad \Delta_f H^{\ominus} = -75\,kJ\,mol^{-1}$

Answer

So, for the following reaction:

$C(g) + 2H_2(g) \rightarrow CH_4(g)$

$\Delta_r H = -75 + (-715) = -790\,kJ\,mol^{-1}$

$\Sigma(\text{bonds broken}) = 2E(H{-}H) = +872\,kJ\,mol^{-1}$

$\Sigma(\text{bonds made}) = 4E(C{-}H)\,kJ\,mol^{-1}$

$\Delta_r H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds made})$

$-790 = +872 - 4E(C{-}H)$

$E(C{-}H) = \frac{1}{4}(790 + 872) = 415.5\,kJ\,mol^{-1}$

BOND ENTHALPIES AND EASE OF REACTION

Bond enthalpies can be used to predict which bonds are most likely to break first in a reaction, and how easy it is to break the bond.

Bonds with high bond enthalpies require more energy to break them. Bonds with relatively low bond enthalpies require less energy to break them. This means that they are easier to break, and therefore are more likely to break first in a chemical reaction.

A reaction that involves breaking bonds with low bond enthalpies is more likely to take place at room temperature than reactions involving molecules with high bond enthalpies. Reactions involving the breaking of bonds with high bond enthalpies are more likely to require heating and/or the use of a catalyst.

STANDARD ENTHALPY CHANGE OF ATOMISATION, $\Delta_{at}H^{\ominus}$

The enthalpy change measured at a stated temperature, usually 298 K, and 100 kPa when one mole of gaseous atoms is formed from an element in its standard state is called the **standard enthalpy change of atomisation** of the element. It is given the symbol $\Delta_{at}H^{\ominus}$.

Equations representing some standard enthalpy changes of atomisation at 298 K are given below.

$C(s) \rightarrow C(g)$ $\quad \Delta_{at}H^{\ominus} = +717\ kJ\ mol^{-1}$

$Na(s) \rightarrow Na(g)$ $\quad \Delta_{at}H^{\ominus} = +107\ kJ\ mol^{-1}$

$\frac{1}{2}H_2(g) \rightarrow H(g)$ $\quad \Delta_{at}H^{\ominus} = +218\ kJ\ mol^{-1}$

$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$ $\quad \Delta_{at}H^{\ominus} = +122\ kJ\ mol^{-1}$

LEARNING TIP

Note that the standard enthalpy change of atomisation is the enthalpy change when one mole of gaseous **atoms** is formed. In the case of elements that exist as polyatomic molecules, it is **not** the enthalpy change when one mole of gaseous **molecules** is atomised.

CHECKPOINT

SKILLS PROBLEM SOLVING

1. Use the following mean bond enthalpies to calculate $\Delta_r H$ for each of the following reactions. Assume that all species are in the gaseous state.

BOND	H–H	C–C	C=C	C–H	C=O	O=O
E / kJ mol^{-1}	436	347	612	413	743	498

BOND	H–O	F–F	C–Br	H–Br	H–F	Br–Br
E / kJ mol^{-1}	464	158	290	366	568	193

(a) $H_2 + F_2 \rightarrow 2HF$

(b) $CH_3CH{=}CH_2 + Br_2 \rightarrow CH_3CHBrCH_2Br$

(c) $2CH_3CH_3 + 7O_2 \rightarrow 4CO_2 + 6H_2O$

(d) $CH_2{=}CH_2 + HBr \rightarrow CH_3CH_2Br$

2. (a) Write an equation to represent the standard enthalpy change of formation of ammonia.

(b) Calculate the mean bond enthalpy for the N–H bond in ammonia using this data:

$\Delta_f H^{\ominus}[NH_3(g)] = -46\ kJ\ mol^{-1}$

$E(N{\equiv}N) = 945\ kJ\ mol^{-1}$

$E(H{-}H) = 436\ kJ\ mol^{-1}$

3. The equation for the formation of sulfur hexafluoride is:

$S(s) + 3F_2(g) \rightarrow SF_6(g)$

Use the following data to calculate the mean bond enthalpy for the S–F bond in sulfur hexafluoride.

$\Delta_f H^{\ominus}[SF_6(g)] = -1100\ kJ\ mol^{-1}$

$E(F{-}F) = 158\ kJ\ mol^{-1}$

$\Delta_{at}H^{\ominus}[S(s)] = +223\ kJ\ mol^{-1}$

SUBJECT VOCABULARY

standard enthalpy change of atomisation the enthalpy change measured at a stated temperature, usually 298 K, and 100 kPa when one mole of gaseous atoms is formed from an element in its standard state